

AS • Edexcel • Physics

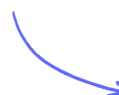
 3 mins  3 questions

Multiple Choice Questions

Resistance, Resistivity & Potential Dividers

Electrical Resistivity / Core Practical 2: Investigating Resistivity / Current & Drift Velocity / Potential Difference & Conductor Length / Potential Dividers / Potential Dividers & Variable Resistance

Scan here to return to the course
or visit [savemyexams.com](https://www.savemyexams.com)



Total Marks

/3

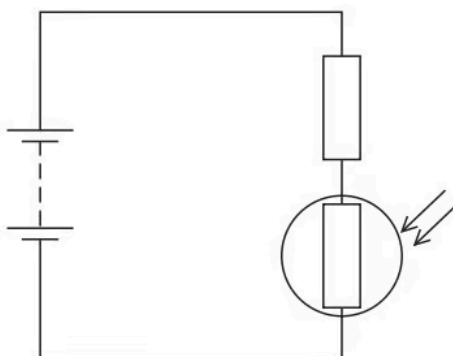
- 1 The intensity of light incident on a light dependent resistor (LDR) can vary both its electrical resistance R and the number of charge carriers per unit volume n . The light intensity on an LDR is increased.

Which row of the table describes the effect on R and n ?

	R	n
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

(1 mark)

- 2 A light dependent resistor (LDR) and a resistor are connected to a battery, as shown.



The intensity of light incident on the LDR increases.

Which row of the table describes the change in the resistance of the LDR and the change in the potential difference across the resistor?

	Resistance of LDR	Potential difference across the resistor
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

(1 mark)

3 A light dependent resistor is connected across a cell of negligible internal resistance.

The light intensity is increased.

Which of the following statements about the current is correct?

- A.** It decreases because there is an increase in the number of conduction electrons.
- B.** It increases because there is an increase in the number of conduction electrons.
- C.** It decreases because the amplitude of lattice vibrations decreases.
- D.** It increases because the amplitude of lattice vibrations increases.

(1 mark)